Series and parallel: where the gauge insertion law stops closing

Halve one lattice direction of a heat-kernel lattice gauge theory in D dimensions. The step splits exactly into two kinds of move. Faces containing the refined direction are cut in two; these *series* moves integrate out exactly in every dimension, by the heat-kernel convolution. Faces transverse to the refined direction acquire a new parallel copy at doubled heat time; integrating out these *parallel* insertions leaves the factor Ψ of Proposition 1, the partition function of a $(D-1)$-dimensional gauge theory on the inserted mid-plane whose edges carry Brownian-bridge laws fixed by the coarse field. There are $\binom{D-1}{2}$ transverse planes: none in $D=2$, which is why two-dimensional Yang–Mills closes, one in $D=3$, three in $D=4$.

What is proved about Ψ:

- *Free field* (Proposition 2): the defect against the coarse action is an exact negative semidefinite quadratic form, of relative size at most $a^2K^2/16 + (Ga)^2/8$ on fields with transverse momenta up to K and layer gradient G. It is the counterpart of Newton's $-F^2uvh/(8M)$ (refinement note, eq. (1)) and exactly the error of Migdal's bond moving.
- *Compact* $U(1)$ (Propositions 3–4, Theorem 5): exact formulas for a cube and a mid-plane, and a proof that one step equals the free step up to an extensive error of density $e^{-\pi^2/(8\lambda_3 a)}$ on the small-field set. This is a one-step estimate for the unperturbed action; stability under iteration remains open. The known massless fixed-coupling and massive fixed-Debye-mass limits require their separate continuum theorems.
- *Non-abelian groups* (Propositions 6–7): the bridge midpoint's curvature term is explicit for every compact group (for $SU(3)$ the averaged softening per cut face is $t_j|X_j|^2/512$), and the one-step inequality in Hypothesis P(α) of §4 holds for one cube conditional on the full normalized Laplace bounds in Proposition 7. Uniformity and stability under iteration remain to be established.

This answers the stop rule of the refinement note, §8, in its second form: it gives the exact term that prevents closure, with an explicit free-field estimate, and states the interacting estimate that remains. The ingredients are established (heat-kernel convolution, Brownian bridges, Gaussian integration); the decomposition and the exact free-field defect are elementary, and no novelty is claimed for them beyond their use here as the organizing comparison. Migdal's recursion (1975) is the approximate form of the same series/parallel split, as Kadanoff's notes describe it (Kadanoff 1976, metadata).

1. Heat times on an anisotropic lattice

Take a hypercubic lattice in D dimensions with spacings $a_1, \dots, a_D$, compact gauge group G with the group metric of the refinement note §5 ($-\Delta_G \chi_R = C_2(R)\chi_R$, $C_2(\mathbf{3}) = 4/3$ for $SU(3)$), and heat kernel $k_t = e^{t\Delta_G/2}$, $k_t(U) = \sum_R d_R \chi_R(U) e^{-tC_2(R)/2}$. Use the dimensionally explicit continuum action of the refinement note §6,

$$\frac{S_E}{\hbar} = \frac{1}{4\lambda_D} \int F^a_{\mu\nu} F^a_{\mu\nu}\, d^D x, \qquad [\lambda_D] = \text{length}^{D-4},$$

so that $\lambda_D = \hbar g^2_{\text{cl},D}$ contains $\hbar$. The anisotropic heat-kernel action assigns to a plaquette p in the plane $(\mu\nu)$ the weight $k_{t_{\mu\nu}}(U_p)$ with

$$t_{\mu\nu} = \lambda_D \, \frac{a_\mu a_\nu}{\prod_{\rho \neq \mu,\nu} a_\rho}. \tag{1}$$

Derivation of (1). For a smooth field, $U_p = \exp(a_\mu a_\nu F_{\mu\nu} + O(a^3))$ and, for small t away from the cut locus, $-\log k_t(e^X) = |X|^2/(2t) + \text{const} + O(t + |X|^2)$. Matching $a_\mu^2 a_\nu^2 |F_{\mu\nu}|^2/(2t_{\mu\nu})$ to the continuum density $(1/2\lambda_D)|F_{\mu\nu}|^2 \prod_\rho a_\rho$ (the factor $1/2$ collects $F_{\mu\nu}$ and $F_{\nu\mu}$) gives (1). □

The heat time is dimensionless. In $D = 2$ it is λ_2 times the face area, as in the refinement note; in $D = 3$ the isotropic value is $t = \lambda_3 a$, the refinement parameter of that note's table; in $D = 4$ it is $t = \lambda_4 = g^2_{\text{lat}}$, independent of a. Equation (1) is a choice of regularization, the heat-kernel counterpart of the anisotropic Wilson action; the theorems below concern it exactly. The matching holds in the small-field regime; under large anisotropy, such as refining time alone, the transverse heat times grow and the Hamiltonian limit requires exponential (Wilson-type) weights instead (zero-spacing note, §1). The halvings used here change heat times by factors of two.

2. Halving one direction: series moves and a parallel insertion

Halve direction 1: $a_1 \mapsto a_1/2$. By (1),

- a plaquette in a plane (1ν) has $t_{1\nu} \mapsto t_{1\nu}/2$, and each coarse (1ν) face is cut by a new edge into two fine faces;
- a plaquette in a plane (jk) with $j, k \neq 1$ has $t_{jk} \mapsto 2t_{jk}$; the old (jk) faces survive with doubled heat time, and a new (jk) face appears in each mid-plane $x_1 = (\ell + \frac{1}{2})a_1$, also at heat time $2t_{jk}$.

The new variables are the half-edges in direction 1 and the *mid-edges* lying in the mid-planes. Each mid-edge in direction $j \neq 1$ lies in exactly one coarse face of the plane $(1j)$, which it cuts in two, and it borders $2(D-2)$ mid-faces.

Proposition 1 (exact pushforward). Let μ' be the fine heat-kernel measure and p the projection multiplying half-edges. Then $p_*\mu'$ has density, with respect to the product Haar measure on coarse links,

$$\frac{1}{Z'} \prod_{f \ni 1} k_{t_f}(U_f) \prod_{f \perp 1} k_{2t_f}(U_f)\ \Psi(U), \tag{2}$$

where $f \ni 1$ ranges over coarse faces containing direction 1, $f \perp 1$ over the others, and

$$\Psi(U) = \int \prod_{g\ \text{mid}} k_{2t_g}(m_{\partial g}) \prod_{e\ \text{mid}} \beta_e(dm_e \,|\, U). \tag{3}$$

Here $m_{\partial g}$ is the ordered product of mid-edge variables around the mid-face g, and β_e is the midpoint law of the Brownian bridge on G of duration t_{1j} determined by the coarse face that e cuts: if the two halves of that face have holonomies $P_e m_e^{-1}$ and $m_e Q_e$, with $P_e Q_e = U_f$,

$$\beta_e(dm \,|\, U) = \frac{k_{t_{1j}/2}(P_e m^{-1})\, k_{t_{1j}/2}(m Q_e)}{k_{t_{1j}}(U_f)}\, dm. \tag{4}$$

Proof. At each new vertex on a direction-1 edge, only the two half-edges and the incident mid-edges meet. Changing variables to (coarse product, upper half) and using Haar invariance, the gauge transformation at that vertex sets the upper half-edge to the identity without changing the integral; the lower half-edge becomes the coarse link. The half-faces then have the holonomies displayed, with P_e, Q_e built from coarse links only. Multiply and divide by $k_{t_{1j}}(U_f)$ for each cut face. By the convolution identity, eq. (9) of the refinement note, each β_e is a probability measure. Fubini gives (2)–(3). □

The dimension count. Refining one direction produces parallel insertions in the $\binom{D-1}{2}$ planes transverse to it: zero for $D = 2$, one for $D = 3$, three for $D = 4$. With no transverse plane, $\Psi = 1$ and (2) is the exact consistency (7) of the refinement note, recovering its §5. An isotropic dyadic step is D successive directional halvings. By (1) the heat times change as follows:

D	one directional halving	full isotropic step	parallel planes per halving
2	$t \mapsto t/2$	$t \mapsto t/4$ (area)	0
3	$t_{1\nu}/2$, $2t_{23}$	$t \mapsto t/2$	1
4	$t_{1\nu}/2$, $2t_{jk}$	$t \mapsto t$	3

The last column locates the whole renormalization problem. The middle column is the classical scaling: superrenormalizable in $D = 3$, where the heat time halves per step, and marginal in $D = 4$, where it returns to itself and any running must come from Ψ.

The structure matches Newton's insertion. There the new time splits one cell in series and the elimination leaves the endpoint-independent scalar $-F^2uvh/(8M)$; here the series moves close with no remainder (the heat kernel is already the exact propagator, the counterpart of U_h rather than Q_h), and the parallel insertion leaves Ψ. A Wilson action would also leave series remainders, as Q_h does.

3. The free field: the parallel defect in closed form

Take $G = \mathbb{R}^n$ with the Euclidean metric (the Gaussian limit of a compact group of dimension n; $n = 8$ for $SU(3)$) and $D = 3$. Then $k_t(x) = (2\pi t)^{-n/2}e^{-|x|^2/(2t)}$, and β_e is Gaussian with mean $(P_e - Q_e)/2$ in additive notation and covariance $(t_{1j}/4)\, I_n$, independent of the coarse data. Put the lattice on a torus with $N_2 \times N_3$ transverse sites and N_1 layers. Write $\Phi_\ell(x)$ for the flux of the old (23) face at layer ℓ and transverse site x, and $\hat\Phi_\ell(k)$ for its unitary discrete Fourier transform in (x_2, x_3), with $s_j = \sin(k_j a_j/2)$ and

$$v(k) = t_{12}\, s_3^2 + t_{13}\, s_2^2.$$

On a torus the transverse zero mode of a lattice curl vanishes, so $k \neq 0$ throughout. All statements concern gauge-invariant functions, which depend on the links only through the fluxes; the non-compact gauge orbits drop out with the normalization constants.

Proposition 2 (exact free-field parallel defect). In this Gaussian theory, the pushforward (2) is the coarse heat-kernel measure multiplied by $e^{-\mathcal{D}}$ (up to a constant), with

$$\mathcal{D} = -\sum_\ell \sum_{k\neq 0} \left[\frac{v(k)\big(|\hat\Phi_\ell|^2 + |\hat\Phi_{\ell+1}|^2\big)}{8t_{23}\big(2t_{23} + v(k)\big)} + \frac{|\hat\Phi_\ell - \hat\Phi_{\ell+1}|^2}{8\big(2t_{23} + v(k)\big)} \right] \;\leq\; 0. \tag{5}$$

Proof. (i) The mean mid-edge configuration interpolates linearly between layers ℓ and $\ell + 1$; the direction-1 links enter it as a lattice gradient, which the curl annihilates. Hence the mean mid-face flux is exactly $(\Phi_\ell + \Phi_{\ell+1})/2$. (ii) The fluctuation of the mid-face flux is the lattice curl of independent edge noises with variances $t_{12}/4$ (direction 2) and $t_{13}/4$ (direction 3). Its Fourier covariance is $(t_{12}/4)|1 - e^{ik_3a_3}|^2 + (t_{13}/4)|1 - e^{ik_2a_2}|^2 = v(k)$, diagonal in k. (iii) For Gaussian η of variance v, $E\, e^{-|\bar\Phi+\eta|^2/(2\sigma^2)} \propto e^{-|\bar\Phi|^2/(2(\sigma^2+v))}$; with $\sigma^2 = 2t_{23}$, $-\log\Psi = \sum_{\ell,k} |\hat\Phi_\ell + \hat\Phi_{\ell+1}|^2/(8(2t_{23} + v)) + \text{const}$. (iv) The old (23) faces contribute $\sum |\hat\Phi_\ell|^2/(4t_{23})$, the coarse target is $\sum |\hat\Phi_\ell|^2/(2t_{23})$, and the cut faces return $k_{t_{1j}}(U_f)$ exactly. Subtract, and use $|A + B|^2 = 2|A|^2 + 2|B|^2 - |A - B|^2$. □

At zero transverse momentum and constant flux in ℓ, the two old and new faces at heat time $2t_{23}$ combine in parallel to one face at t_{23}, as conductances do. The defect measures the two departures: transverse lattice momentum ($v > 0$) and variation between layers.

Corollary (size of one step). On an isotropic coarse lattice ($a_j = a$, $t_{\mu\nu} = t = \lambda_3 a$), suppose $\hat\Phi_\ell$ is supported on $|k| \leq K$ and $\sum_\ell \|\Phi_\ell - \Phi_{\ell+1}\|^2 \leq (Ga)^2 \sum_\ell \|\Phi_\ell\|^2$, with $\|\cdot\|$ the ℓ^2 norm over transverse sites. Then

$$|\mathcal{D}| \le \Big(\frac{a^2K^2}{16} + \frac{(Ga)^2}{8}\Big) \sum_\ell \frac{\|\Phi_\ell\|^2}{2t}. \tag{6}$$

Proof. Use $2t + v \ge 2t$ and $v \le ta^2|k|^2/4$ in (5). □

Corollary 2_s (a cut at fraction s). Cut every direction-1 cell at $x_1 = (\ell+s)a_1$, $0 < s < 1$, and give each transverse face the heat time $\lambda_3 a_2 a_3/\ell_1$ with ℓ_1 its dual length in direction 1 (the trapezoid weight). The two layers adjacent to any old or new (23) face are a distance a_1 apart, so every transverse face has dual length $a_1/2$ and keeps heat time $2t_{23}$; the cut faces have st_{1j} and $(1-s)t_{1j}$, and the mid-edge law is the Gaussian bridge at fraction s, with mean $(1-s)P_e - sQ_e$ (in the additive notation above, where $s = \frac{1}{2}$ gives $(P_e - Q_e)/2$, and with $P_e m_e^{-1}$ the holonomy of the half of thickness sa_1) and covariance $s(1-s)t_{1j}I_n$. Steps (i)–(iv) of the proof go through with mean mid-edge $(1-s)A_j(\ell) + sA_j(\ell+1)$ plus a transverse gradient, hence mean mid-face flux $(1-s)\Phi_\ell + s\Phi_{\ell+1}$, and with noise covariance $v_s = 4s(1-s)v(k)$. Using $|(1-s)A + sB|^2 = (1-s)|A|^2 + s|B|^2 - s(1-s)|A-B|^2$ and periodicity in ℓ,

$$\mathcal{D}_s = -\sum_\ell \sum_{k\neq 0} \left[\frac{v_s(k)\,|\hat\Phi_\ell|^2}{4t_{23}\big(2t_{23} + v_s(k)\big)} + \frac{s(1-s)\,|\hat\Phi_\ell - \hat\Phi_{\ell+1}|^2}{2\big(2t_{23} + v_s(k)\big)} \right] \le 0,$$

which is (5) at $s = \frac{1}{2}$. Under the hypotheses of the Corollary, $|\mathcal{D}_s| \le 4s(1-s)(\frac{a^2K^2}{16} + \frac{(Ga)^2}{8}) \sum_\ell \|\Phi_\ell\|^2/(2t)$: the bound (6) times $4s(1-s)$, so halving is the largest single step and an off-centre cut is gentler in both the noise and the displacement parts; it also shrinks the mesh less, to $\max(s, 1-s)\,a_1$. The trapezoid weights are the diagonal Hodge star of discrete exterior calculus on a tensor grid (dual over primal volume), so the independence of the free-field limit from the cut sequence is a convergence question for that discretization; its proofs so far cover the Hodge–Laplace problem in two dimensions on shape-regular Delaunay meshes (Zhu, Christiansen, Hu and Hirani, arXiv:2505.08966, abstract), and for nested tensor grids in $D = 3, 4$, with bounded aspect ratios as the natural hypothesis, it remains expected. The statement covers one step from a uniform lattice; after it the layers have thicknesses sa_1 and $(1-s)a_1$, and the next cut starts from that non-uniform lattice. Theorem 5 for a general cut is Corollary 5_s below. □

For a smooth field, $\Phi = a^2 F_{23}$ and the right side of (6) is $a^2(K^2/16 + G^2/8)$ times the continuum magnetic action $(1/2\lambda_3)\int |F_{23}|^2 d^3x$: a dimension-six correction with coefficient a^2. At fixed physical K and G the relative defects along $a_n = 2^{-n}a_0$ are summable, which is the free-field instance of the refinement note's criterion (12), in a norm on Gaussian actions rather than total variation. Two cautions keep this honest. After one step the effective action is (2) with $\mathcal{D}$, so the next step starts from a slightly different action; the free field is controlled exactly because every stage is Gaussian, and its blocked actions should converge to a Gaussian fixed point. That fixed point goes back to Bell and Wilson (1975) (metadata), and the perfect-action programme of Hasenfratz and Niedermayer (1994) (metadata) develops it for asymptotically free models with a smeared blocking; for the product blocking used here the convergence is asserted, not proved. And (6) holds for smooth coarse data; typical lattice fields at fixed t are rough, so (6) states the irrelevance of the defect, while the measure-level estimate needs the whole quadratic form (5).

Migdal's bond moving, read exactly. Moving each mid-face onto the old face of layer ℓ multiplies two Gaussian weights at $2t_{23}$ into one at t_{23}, which is the coarse target. So in the Gaussian theory $\mathcal{D}$ is exactly the error of the bond-moving step, and (5) splits it into its noise part (v) and its displacement part ($\Phi_\ell - \Phi_{\ell+1}$). Tomboulis's attempt to bound the exact decimation between Migdal–Kadanoff-type approximations (arXiv:0707.2179, abstract; not established as a theorem, see what-would-unblock) worked on the same parallel step.

The comparison with Newton. Both eliminations complete a square, and both lower the action. Newton's defect is a constant, removed by one cubic counterterm, after which the

cell law closes exactly. The free-field defect is an irrelevant quadratic form; no finite set of nearest-neighbour counterterms removes it, and the closed object is the fixed point in the space of all quadratic actions. In the interacting theory Ψ is non-Gaussian.

4. The non-abelian parallel move in $1+2$ dimensions

For compact G, (3) says: $\Psi(U)$ **is the partition function of a two-dimensional heat-kernel lattice gauge theory on the mid-planes, at heat time $2t_{23}$, in which each edge is distributed by the bridge law $\beta_e(\cdot\,|\,U)$ instead of Haar measure.** Two limits are exactly soluble. As $t_{1j}\to\infty$ the bridge laws tend to Haar measure, the mid-plane is the exactly soluble two-dimensional theory of the refinement note §5, and Ψ is a constant: the old transverse faces stay at heat time $2t_{23}$. As $t_{1j}\to 0$ the bridge laws concentrate at the geodesic interpolation and Ψ is the product of mid-face weights on the interpolated holonomies, a classical evaluation, like Newton's z_*. Between those limits, two-dimensional exact solvability fails, because it rests on Haar invariance of every edge.

Relative to the Gaussian computation of §3, four terms appear, with the orders that power counting assigns them at $t=\lambda_3 a\ll 1$ (labelled as power counting; proving them is the content of the hypothesis below):

- **N1, interpolation.** The mid-face holonomy of the geodesic interpolation (the $t_{1j}\to 0$ bridge) differs from the abelian average by Baker–Campbell–Hausdorff commutators; in the axial gauge realized by the mid-vertex gauge fixing the leading term is $-\frac{1}{8}a^4[F_{12},F_{13}]$ in $\log U_g$. In the action this is a dimension-six operator, $O(a^2)$ relative, like (6).
- **N2, bridge shape.** The bridge law depends on the coarse face holonomy through the group's curvature (the Jacobian of the exponential map in the small-t heat kernel). Relative size $O(t)=O(\lambda_3 a)$; gauge invariance allows it to shift the coupling and the vacuum energy only, at leading order.
- **N3, non-Gaussian fluctuation.** The mid-face holonomy is a non-commutative product of noisy edges; the connected corrections to $-\log\Psi$ are $O(t)$ relative and local.
- **N4, large fields.** Bridges that wind around the group (the image terms of the heat kernel near the cut locus) contribute $O(e^{-c/t})$ per face for coarse fluxes a fixed distance from the cut locus, and $O(1)$ at it (Proposition 3 below gives the rate); the small-field split of Hypothesis P(α) handles them.

Correction, 2026-09-26 (referee check). The first version of the hypothesis below bounded $-\log\Psi$ itself and demanded large-field suppression of Ψ; the free field violates the first (it needs $\delta_t=\frac{1}{2}$) and Proposition 3 the second. The version below bounds the log-ratio $\mathcal{D}$ of (2) against the coarse density.

Hypothesis P(α) (one parallel step, $D=3$). Fix $\delta\in(0,\frac{1}{4})$. Let

$$\mathcal{D}(U)=-\log\Psi(U)-\sum_{f\perp 1}\log\frac{k_{2t_{23}}(U_f)}{k_{t_{23}}(U_f)},\qquad S_{\mu\nu}(U)=\sum_{f\in(\mu\nu)}\frac{|\log U_f|^2}{2t_{\mu\nu}},$$

so that $p_*\mu'$ is the coarse heat-kernel density times $e^{-\mathcal{D}}$ (for the free field, $\mathcal{D}$ is (5)). There exist $\alpha>2\delta$, $t_0>0$, $C<\infty$, a local quadratic form $\mathcal{D}_0$ in $\{\log U_p\}$ (the form (5), with t_{23} possibly renormalized by $O(t)$), and for each $t=\lambda_3 a\le t_0$ numbers $c_{\mathrm{vac}}(t)$ and coupling shifts $\delta_t^{\mu\nu}$, one per plane, with $|\delta_t^{\mu\nu}|\le Ct$, such that on the small-field set $\{|\log U_p|\le t^{1/2-\delta}\text{ for all }p\}$

$$\left|\mathcal{D}(U)-\mathcal{D}_0(U)-c_{\mathrm{vac}}N_{\mathrm{cells}}-\textstyle\sum_{\mu<\nu}\delta_t^{\mu\nu}S_{\mu\nu}(U)\right|\le C\,t^{\alpha}\textstyle\sum_p(1+|\log U_p|^2/t),$$

the sum running over all coarse plaquettes. The same statement is required for heat-kernel actions perturbed by local terms of the size on the right, so that it can be iterated. No large-field clause on Ψ is needed: $\Psi\le\prod_g k_{2t_g}(1)$, and the explicit factors of (2) suppress each plaquette

outside the small-field set by $e^{-t^{-2\delta}/4}$, up to a factor $t^{-n/2}$ per cell. Free-field check: $\mathcal{D} = \mathcal{D}_0$, all $\delta_t^{\mu\nu} = 0$, every α. (Amended the same night: the first version allowed a shift of the transverse coupling only; the one-loop determinant below shifts the couplings of the cut faces as well.)

(Amended 2026-09-27, from the SU(2) mid-plane note and its referee: on periodic planes $\mathrm{P}(\alpha)$ is required for plane sizes $\min_j N_j \geq c_0/t$, fixed physical size; at fixed N with $t \to 0$ a flat-holonomy winding determinant of the massive mid-plane modes is of order one and violates the literal statement.)

The compact abelian cube isolates N4 exactly. Take $G = U(1)$ with $k_t(\theta) = \sum_{n\in\mathbb{Z}} e^{-tn^2/2} e^{in\theta}$ (the Villain weight), and refine a single cube: one mid-face at heat time t_m, bounded by four mid-edges m_j with orientations $\sigma_j = \pm 1$, each cutting a side face of heat time t_j whose halves have angles $p_j - m_j$ and $m_j + q_j$. Put $\phi_j = p_j + q_j$ (the side-face flux, any real representative), $\bar{\Phi} = \sum_j \sigma_j (p_j - q_j)/2$ (the interpolated mid-face flux, with the same representatives), and

$$\rho_t(\phi) = \frac{\sum_{w\in\mathbb{Z}} (-1)^w e^{-(\phi - 2\pi w)^2/(2t)}}{\sum_{w\in\mathbb{Z}} e^{-(\phi-2\pi w)^2/(2t)}} \in [-1, 1].$$

Proposition 3 (exact $U(1)$ cube). With $T = t_m + \frac{1}{4}\sum_j t_j$,

$$\Psi = \sum_{n \text{ even}} e^{-Tn^2/2} e^{in\bar{\Phi}} + \Big(\prod_{j=1}^{4} \rho_{t_j}(\phi_j)\Big) \sum_{n \text{ odd}} e^{-Tn^2/2} e^{in\bar{\Phi}}. \tag{7}$$

As every $\rho_{t_j}(\phi_j) \to 1$, $\Psi \to k_T(\bar{\Phi})$. For $|\phi_j| \leq \pi$ and $t_j \leq 1$, $0 \leq 1 - \rho_{t_j}(\phi_j) \leq 5\, e^{-2\pi(\pi - |\phi_j|)/t_j}$.

Proof. The bridge numerator for $E\, e^{inm}$ is $\sum_r e^{-t(r^2 + (r-n)^2)/4} e^{irp} e^{i(r-n)q}$ by character orthogonality, and $r^2 + (r-n)^2 = 2(r - n/2)^2 + n^2/2$. For even n the shift $r \mapsto r + n/2$ reproduces the denominator $k_t(\phi)$, giving $E\, e^{inm} = e^{-tn^2/8} e^{in(p-q)/2}$. For odd n the shift is by a half-integer and leaves the ratio $\sum_s e^{-t(s-\frac{1}{2})^2/2} e^{i(s-\frac{1}{2})\phi} / \sum_s e^{-ts^2/2} e^{is\phi}$, which Poisson summation turns into $\rho_t(\phi)$. The four m_j are independent, so $E\, k_{t_m}(\sum_j \sigma_j m_j)$ is the sum over n of $e^{-t_m n^2/2} \prod_j E\, e^{in\sigma_j m_j}$, which is (7). For the bound, write $G_w = e^{-(\phi - 2\pi w)^2/(2t)}$; then $1 - \rho = 2\sum_{w \text{ odd}} G_w / \sum_w G_w \leq 2\sum_{w \text{ odd}} G_w / G_0$. For $|\phi| \leq \pi$ the odd term nearest ϕ gives $e^{-2\pi(\pi - |\phi|)/t}$, the opposite one at most the same, and the terms $|w| \geq 3$ are smaller by $e^{-12\pi^2/t}$; for $t \leq 1$ the total is below $5e^{-2\pi(\pi-|\phi|)/t}$. □

Equation (7) displays the parallel move for one cube with N1–N3 absent (the group is abelian and flat): the mid-face weight is the heat kernel at the *softened* heat time $t_m + \frac{1}{4}\sum t_j$, the isolated-cube counterpart of the noise $v(k)$ in (5) (indeed $\frac{1}{4}\sum_j t_j = (t_{12} + t_{13})/2$ is the Brillouin-zone average of v), evaluated at the interpolated flux. The only departure is the odd-charge factor, which differs from 1 only where a side-face flux approaches the cut locus $|\phi| = \pi$, where its bridge midpoint becomes bimodal. That is N4, exactly, with the rate $e^{-2\pi(\pi - |\phi|)/t}$. At the cut locus itself, $\rho = 0$ and $\Psi = \frac{1}{2}[k_T(\bar{\Phi}) + k_T(\bar{\Phi} + \pi)]$, at most a factor 2 below its small-field value: large side fluxes are suppressed by the explicit factors $k_{t_{1j}}(U_f)$ of (2), and Ψ adds no suppression.

The whole $U(1)$ mid-plane. The same computation extends to a full mid-plane of N mid-faces g and mid-edges e, with δ the coboundary from faces to edges, $(\delta n)_e$ the signed sum of n_g over the two mid-faces at e, t_e the heat time of the side face that e cuts, and

$$C = 2t_{23}\, I + \delta^{\mathsf{T}} \mathrm{diag}(t_e/4)\, \delta,$$

a positive definite $N \times N$ matrix. $\bar{\Phi}_g$ is the interpolated flux of g and ϕ_e the flux of the side face cut by e.

Proposition 4 (exact $U(1)$ mid-plane).

Referee verdict, 2026-09-27: ACCEPT. Equations (8)–(9) and the independent parity-mixture proof have the stated normalized-Haar constants.

$$\Psi = \sum_{n\in\mathbb{Z}^N} e^{-\frac{1}{2}n^{\mathsf{T}}Cn}\, e^{in\cdot\bar\Phi} \prod_{e:\,(\delta n)_e \text{ odd}} \rho_{t_e}(\phi_e). \tag{8}$$

If every $\rho_{t_e}(\phi_e)$ is replaced by 1, Poisson summation gives the vortex form

$$\Psi_1(\bar\Phi) = \frac{(2\pi)^{N/2}}{\sqrt{\det C}} \sum_{w\in\mathbb{Z}^N} e^{-\frac{1}{2}(\bar\Phi+2\pi w)^{\mathsf{T}}C^{-1}(\bar\Phi+2\pi w)}. \tag{9}$$

Proof. The proof of Proposition 3 shows that each bridge midpoint is $m_e = \bar m_e + \pi s_e + \xi_e$ modulo 2π, with $\bar m_e$ the interpolation, $s_e \in \{0,1\}$ with $P(s_e = 1) = (1-\rho_e)/2$, and ξ_e Gaussian of variance $t_e/4$, all independent. (The bridge density is a product of two wrapped Gaussians; its two winding numbers combine into a total, which carries the weight $e^{-(\phi_e+2\pi W)^2/(2t_e)}$, and a relative one, which shifts the centre by πW.) Expand each mid-face weight $k_{2t_{23}}(\cdot) = \sum_{n_g} e^{-t_{23}n_g^2} e^{in_g(\cdot)}$. The Gaussian average gives $e^{-\sum_e t_e(\delta n)_e^2/8}$, and the average over s_e gives 1 for even $(\delta n)_e$ and ρ_e for odd; this is (8). For (9), apply the Poisson summation formula on $\mathbb{Z}^N$. □

For one mid-face, (8) is (7). With isotropic coarse heat times ($t_e = t_{23} = t$), $C = 2t(I + L/8)$ with $L = \delta^{\mathsf{T}}\delta$ the Laplacian of the dual lattice, whose spectrum lies in $[0, 8]$. In Fourier variables $C = 2t + v(k)$, the covariance of Proposition 2, so the $w = 0$ term of (9) is the free field exactly. The vortices $w \neq 0$ are the compact correction, a periodic Gaussian of the Villain type (Banks, Myerson and Kogut 1977, metadata, for the duality in abelian lattice gauge theories).

Corollary (vortex bound). In the isotropic case, write $\Psi_1^{(0)}$ for the $w = 0$ term of (9) and assume $\|\bar\Phi\|_\infty < \pi/2$. Then

$$0 \le \log \frac{\Psi_1(\bar\Phi)}{\Psi_1^{(0)}(\bar\Phi)} \le \frac{2N\, e^{-c/t}}{1 - e^{-c/t}}, \qquad c = \pi\Big(\frac{\pi}{2} - \|\bar\Phi\|_\infty\Big). \tag{10}$$

Referee verdict, 2026-09-27: ACCEPT. The M-matrix argument applies to the periodic square mid-plane used here; taking the logarithm of the product bound gives (10).

Proof. $I + L/8$ is a symmetric M-matrix with row sums 1, so its inverse has nonnegative entries and row sums 1. Hence $\|C^{-1}\|_{\infty\to\infty} = 1/(2t)$, and $C^{-1} \ge 1/(4t)$ because $L \le 8$. For integer w, $\|w\|_2^2 \ge \|w\|_1$. The exponent of the w term minus that of the $w = 0$ term is $-\frac{1}{2}(2\pi w)^{\mathsf{T}}C^{-1}(2\pi w) - 2\pi w^{\mathsf{T}}C^{-1}\bar\Phi \le -\pi^2\|w\|_1/(2t) + \pi\|w\|_1\|\bar\Phi\|_\infty/t = -c\|w\|_1/t$. Every term is positive, and $\sum_{w\neq 0} e^{-c\|w\|_1/t} = (1 + 2e^{-c/t}/(1-e^{-c/t}))^N - 1$. □

The bound is extensive with density $O(e^{-c/t})$ per mid-face, the form of error that Hypothesis P(α) allows, for every α. The parity factors are handled by the same device, which completes the $U(1)$ step.

Correction (GPT-6 Astra referee), 2026-09-27 — REFINE: use coarse-cube Bianchi identities for the lifts, retain harmonic flux sectors, and claim only the unperturbed one-step part of P(α).

Theorem 5 (unperturbed $U(1)$ one-step estimate). Take $G = U(1)$, $D = 3$, an isotropic coarse lattice with $t = \lambda_3 a$, one halving of direction 1, and a coarse configuration in the small-field set: every coarse plaquette angle, taken in $(-\pi, \pi]$, has modulus at most $\varepsilon = t^{1/2-\delta}$, with $3\varepsilon \le \pi/8$ and $e^{-\pi^2/(16t)} \le \frac{1}{3}$. Use these principal values for the coarse fluxes and choose the interpolated flux representative modulo 2π as $\bar\Phi_g = (\Phi_\ell + \Phi_{\ell+1})_g/2$. Then

$$-10\, N\, e^{-2\pi(\pi-\varepsilon)/t} \;\le\; \log\Psi - \log\Psi_1^{(0)}(\bar\Phi) \;\le\; \frac{2N\, e^{-\pi^2/(8t)}}{1 - e^{-\pi^2/(8t)}},$$

where $\Psi_1^{(0)}(\bar\Phi) = (2\pi)^{N/2}(\det C)^{-1/2} e^{-\frac{1}{2}\bar\Phi^{\mathsf{T}}C^{-1}\bar\Phi}$ is exactly the free-field parallel Gaussian integral. On the sector with zero total principal transverse flux in each periodic layer, $\mathcal{D}$ equals (5) plus a constant, up to an error at most $C_0 N e^{-\pi^2/(8t)}$ with C_0 absolute. On other sectors use

the same real quadratic calculation including its harmonic ($k = 0$) flux. Thus the unperturbed estimate in P(α) holds with $\delta_t = 0$ for every α; its perturbed-action stability clause requires a further proof.

Proof. (i) Representatives. Formula (8) is 2π-periodic in each $\bar\Phi_g$ and invariant under the joint change of lifts ($\phi_e \mapsto \phi_e + 2\pi$ flips ρ_e and shifts the adjacent $\bar\Phi_g$ by π), so any consistent lifts may be used. Take principal ϕ_e by changing the midpoint lifts together with the parity weights. The boundary identity for a coarse cube gives $\sum_{e\in\partial g}\sigma_e\phi_e = \Phi_{\ell+1} - \Phi_\ell + 2\pi n$. The absolute value of the difference of the six principal face terms is at most $6\varepsilon < 2\pi$, hence $n = 0$. Since $\bar\Phi_g \equiv \Phi_{\ell+1} - \frac12\sum_e \sigma_e\phi_e \pmod{2\pi}$, its small representative is the asserted average and has modulus at most ε. This uses only coarse data; the fluctuating half-cell faces have no assumed smallness. Global torus flux can still be nonzero. (ii) Mixture form. By the proof of Proposition 4, $\Psi = E_s\,\Psi_1(\bar\Phi+\pi\delta^{\mathsf T}s)$ with independent $s_e \in \{0,1\}$, $P(s_e = 1) = q_e = (1-\rho_e)/2 \le \frac52 e^{-2\pi(\pi-\varepsilon)/t}$ by Proposition 3. (iii) Lower bound. Every Ψ_1 is positive, so $\Psi \ge \prod_e(1-q_e)\,\Psi_1(\bar\Phi) \ge \prod_e(1-q_e)\,\Psi_1^{(0)}(\bar\Phi)$; there are $2N$ mid-edges and $\log(1-q) \ge -2q$ for $q \le \frac12$. (iv) Upper bound, uniform in s. For $c = \delta^{\mathsf T}s$, the terms of (9) at $\bar\Phi+\pi c$ are indexed by $u = c+2w$, the integer vectors congruent to c modulo 2. Relative to the $w = 0$ term at $\bar\Phi$ the exponent is $-\pi u^{\mathsf T}C^{-1}\bar\Phi - \frac12\pi^2 u^{\mathsf T}C^{-1}u \le -c'\|u\|_1/t$, with $c' = \frac{\pi}{2}(\frac{\pi}{4} - \|\bar\Phi\|_\infty) \ge \pi^2/16$, by the two matrix facts of the Corollary and $\|u\|_2^2 \ge \|u\|_1$. Summing coordinatewise, $\Psi_1(\bar\Phi+\pi c)/\Psi_1^{(0)}(\bar\Phi) \le A_0^{N-k}A_1^k \le A_0^N$, where k is the number of odd entries of c, $A_0 = 1 + 2e^{-2c'/t}/(1-e^{-2c'/t})$ and $A_1 = 2e^{-c'/t}/(1-e^{-2c'/t}) \le A_0$ under the stated condition on t. Average over s and use $\log A_0 \le 2e^{-2c'/t}/(1-e^{-2c'/t})$. In fact, writing $r = e^{-c'/t}$, $A_0 - A_1 = (1-r)/(1+r) \ge 0$ for every $t > 0$; the uniform-in-s bound needs no further restriction. (v) The old transverse faces. $\log[k_{2t}(\theta)/k_t(\theta)]$ for Villain weights equals $\theta^2/(4t) + \frac12\log\frac12$ up to the one-dimensional vortex corrections, at most $3e^{-2\pi(\pi-\varepsilon)/(2t)}$ per face for $|\theta| \le \varepsilon$ by the same Poisson estimate; this is where $e^{-\pi^2/(8t)}$ dominates. Collect (iii)–(v) with the definition of $\mathcal{D}$. □

This is the first rung of the ladder completed as a theorem: for the compact abelian group, one directional step of the three-dimensional refinement equals the free-field step of Proposition 2 up to an extensive error of density $e^{-\pi^2/(8\lambda_3 a)}$ on the small-field set. It concerns the heat-kernel action; iterating it requires the same estimate for heat-kernel actions perturbed by the defect, which is the stability clause of P(α). The non-abelian terms N1–N3 are absent here, so Theorem 5 tests the large-field and representative bookkeeping, and leaves the curvature and commutator terms to the $SU(2)$ cube.

Corollary 5$_s$ (compact $U(1)$, a cut at fraction s). Take the hypotheses of Theorem 5 with the trapezoid weights of Corollary 2$_s$, a cut at fraction $s \in (0,1)$, and the representative $\bar\Phi = (1-s)\Phi_\ell + s\Phi_{\ell+1}$. Put $c_s^* = \pi^2/(1+4s(1-s)) \in [\pi^2/2, \pi^2]$, $r_s = e^{-(c_s^*-\pi\varepsilon)/t}$, $h_* = \varepsilon^2(1+4s(1-s))/(4t)$, and let $q_e \le \frac52 e^{-2\pi(\pi-\varepsilon)/t}$ be the probability that the total winding W_e of mid-edge e is nonzero. Then

$$-\sum_e 2q_e \;\le\; \log\Psi - \log\Psi_1^{(0)}(\bar\Phi) \;\le\; \frac{2Nr_s}{1-r_s} + \sum_e q_e S_*^2, \qquad S_* = e^{h_*} + e^{-(c_s^*/2-\pi\varepsilon)/(2t)} + \frac{2r_s}{1-r_s},$$

with $\Psi_1^{(0)}$ built from the covariance C_s of (b) below. For $\varepsilon \le \pi/24$ (Theorem 5's hypothesis) both error densities are $O(e^{-\pi^2/(8t)})$ for every s, so Theorem 5 holds for every cut fraction, and $\mathcal{D}$ equals $\mathcal{D}_s$ of Corollary 2$_s$ plus a constant up to such an error on the sector of zero total transverse flux. At $s = \frac12$ the vortex term $2Nr_{1/2}/(1-r_{1/2})$, $r_{1/2} = e^{-(\pi^2/2-\pi\varepsilon)/t}$, sharpens step (iv) of Theorem 5, whose bound uniform in the parities was lossy.

Proof. (a) *Midpoint law.* The bridge density is

$$\sum_{w_1,w_2} \exp\left[-\frac{(P_e - m + 2\pi w_1)^2}{2st_e} - \frac{(m + Q_e + 2\pi w_2)^2}{2(1-s)t_e}\right]$$

. With $W = w_1 + w_2$ fixed, the minimum over m is $(\phi_e + 2\pi W)^2/(2t_e)$, $\phi_e = P_e + Q_e$, the curvature is $1/(s(1-s)t_e)$, the prefactor is $(2\pi t_e)^{-1/2}$ for every s, and the centre is $\bar{m}_e + 2\pi(1-s)W - 2\pi w_2$; the last term vanishes modulo 2π. So the midpoint is a mixture over $W \in \mathbb{Z}$, with weights $\propto e^{-(\phi_e+2\pi W)^2/(2t_e)}$ independent of s, of wrapped Gaussians of variance $s(1-s)t_e$ centred at $\bar{m}_e + 2\pi(1-s)W$. At $s = \frac{1}{2}$ the shifts are πW, the parity mixture of the proof of Proposition 4, and q_e obeys Proposition 3's bound for every s. (b) *Covariance.* The Gaussian average of that proof now gives $e^{-\frac{1}{2}s(1-s)t\, n^{\mathsf{T}} L n}$, so in the isotropic case (9) has covariance $C_s = 2t(I + \frac{1}{2}s(1-s)L)$, a symmetric diagonally dominant M-matrix with row sums 1: $\|C_s^{-1}\|_{\infty\to\infty} = 1/(2t)$, and $C_s^{-1} \geq 1/(2t(1+4s(1-s)))$ since $L \leq 8$. (The vortex bound (10) holds with $c_s = \pi(\pi/(1+4s(1-s)) - \|\bar{\Phi}\|_\infty) \geq c$.) (c) *Representative.* The mean mid-edges are $(1-s)A_j(\ell) + sA_j(\ell+1)$ plus a transverse gradient, and the argument of Theorem 5 (i) with weights $1-s$ and s gives mean mid-face flux $\bar{\Phi}$ up to multiples of 2π, with $|\bar{\Phi}| \leq \varepsilon$. (d) *Upper bound.* $\Psi = E_W\, \Psi_1(\bar{\Phi} + 2\pi(1-s)\delta^{\mathsf{T}} W)$. For given W the terms of (9) are indexed by $u = (1-s)\delta^{\mathsf{T}} W + w$, $w \in \mathbb{Z}^N$, and relative to the $w = 0$ term at $\bar{\Phi}$ the exponent is $F(u) = -2\pi u^{\mathsf{T}} C_s^{-1}\bar{\Phi} - 2\pi^2 u^{\mathsf{T}} C_s^{-1} u \leq \sum_g h(|u_g|)$, $h(x) = (\pi\varepsilon x - c_s^* x^2)/t$, by the two matrix facts of (b). Here $h \leq h_*$, and $h(x) \leq -(c_s^* - \pi\varepsilon)x/t$ for $x \geq 1$. Summing over w coordinatewise, $\Psi_1(\bar{\Phi} + 2\pi(1-s)\delta^{\mathsf{T}} W)/\Psi_1^{(0)}(\bar{\Phi}) \leq \prod_g S(\theta_g)$, θ_g the fractional part of u_g, $S(\theta) = \sum_k e^{h(|\theta+k|)}$, with $S(0) \leq 1 + 2r_s/(1-r_s)$ and $S(\theta) \leq S_*$ (one point of $\theta + \mathbb{Z}$ within $\frac{1}{2}$ of 0, one in $[\frac{1}{2}, 1)$, the rest at distance at least 1). Only the two mid-faces adjacent to an edge with $W_e \neq 0$ can have $\theta_g \neq 0$, and $S_* \geq S(0) \geq 1$, so $\prod_g S(\theta_g) \leq S(0)^N S_*^{2\#\{e:W_e\neq 0\}}$. The W_e are independent, so $E_W\, S_*^{2\#\{e:W_e\neq 0\}} \leq \prod_e (1 + q_e S_*^2)$; take logarithms. When $\min(s, 1-s) > \varepsilon(1+4s(1-s))/\pi$ the shifted terms are in fact suppressed, and the factor e^{h_*} enters only for cuts within about ε of an end. (e) *Lower bound.* Every term is positive, so $\Psi \geq \prod_e (1-q_e)\Psi_1^{(0)}(\bar{\Phi})$, and $\log(1-q) \geq -2q$. (f) *Old faces and sizes.* Step (v) of Theorem 5 is unchanged, since every transverse face keeps heat time $2t$. For $\varepsilon \leq \pi/24$: $c_s^* - \pi\varepsilon \geq 11\pi^2/24$, and $2\pi(\pi - \varepsilon) - 2th_* \geq 2\pi^2 \cdot \frac{23}{24} - \frac{\pi^2}{576} > \pi^2/8$, although $h_* = \varepsilon^2/(4t) \cdot (1 + 4s(1-s))$ grows like $t^{-2\delta}$. □

What survives: compact $U(1)$ in three dimensions. This theory is the proved example of a lattice gap that the fixed-coupling refinement removes. Göpfert and Mack prove a nonzero string tension for all couplings of the Villain theory, bounded below through a Debye mass $m_D > 0$ (the monopole mechanism of Polyakov 1977, metadata), and establish a continuum limit $a \to 0$ at fixed physical m_D that is a free scalar field of mass m_D, with string tension divided by m_D^2 diverging in that limit (Göpfert–Mack 1982, abstract as indexed; publisher abstract checked 2026-09-27). Gross proves that at fixed physical coupling the Villain theory converges to the massless free electromagnetic field; his Wilson-action statement concerns the electric sector (Gross 1983, publisher abstract).

Correction (GPT-6 Astra referee), 2026-09-27 — REFINE: the source results have different scalings and sectors; a one-step upper bound alone establishes neither continuum convergence nor the monopole density.

In a fixed physical box the Theorem 5 bound is of order $(L/a_n)^2 e^{-\pi^2/(8t_n)}$ per mid-plane and $(L/a_n)^3 e^{-\pi^2/(8t_n)}$ over all planes. At $t_n = \lambda_3 a_n$ its tail vanishes and is summable. Turning this into convergence needs iteration stability and large-field probability control. Göpfert–Mack's fixed-m_D trajectory instead has logarithmically decreasing t_n; their Debye formula supplies its normalization. The bound's constant $\pi^2/8$ cannot determine the error's actual order or the monopole density on that trajectory.

In four dimensions Driver 1987 (publisher abstract) proves convergence on the current sector to a renormalized free electromagnetic field for general energy functions at sufficiently large inverse coupling. For the Wilson energy he obtains convergence at arbitrary coupling by an appropriate Gibbs-state choice, and state independence away from at most countably many coupling values. The sector and state qualifications matter. These results motivate the comparison with the expected nonabelian three-dimensional gap $C_3\hbar c\lambda_3$; locating that gap in particular

N1–N3 terms needs infrared control beyond the present ultraviolet estimate.

For $SU(2)$ and $SU(3)$ the bridge expectations $E\,R(m_j)$ are matrices, and the isolated cube is where N2 and N3 first appear.

The $SU(2)$ bridge midpoint. In the group metric of §1, $SU(2)$ is the three-sphere of radius 2 (curvature $\frac{1}{4}$), with cut locus at geodesic distance 2π and $|\rho|^2 = \frac{1}{4}$. Poisson summation of the character series gives the exact kernel

$$k_t(e^X) = e^{t/8}\sqrt{8\pi}\,t^{-3/2}\sum_{w\in\mathbb{Z}}\frac{\theta-4\pi w}{\sin(\theta/2)}\,e^{-(\theta-4\pi w)^2/(2t)}, \qquad \theta = |X|, \tag{11}$$

with respect to normalized Haar measure; the $w=0$ term is the Gaussian with the Jacobian $(\theta/2)/\sin(\theta/2)$, and $e^{t/8} = e^{t|\rho|^2/2}$. For a cut side face, left-translate so that the bridge runs from e to e^X, $d = |X| < 2\pi$, and write the midpoint as $m = m_* e^{\xi}$ with $m_* = e^{X/2}$.

Proposition 6 ($SU(2)$ midpoint: structure and leading covariance).

Correction (GPT-6 Astra referee), 2026-09-27 — REFINE: the covariance is accepted; the representation formula is a fixed-representation expansion through order t, and the logarithm uses its almost-everywhere principal branch.

(a) Exactly, $E\,D^J(m) = D^J(m_*)\Lambda^J$ with Λ^J real and diagonal in the weight basis of the axis $\hat{X}$, and all odd moments of ξ vanish. (b) As $t\to 0$, uniformly for $d \le d_0 < 2\pi$, ξ is Gaussian to leading order with covariance

$$\frac{t}{4}\,\mathrm{diag}\Big(1, \frac{1}{h(d)}, \frac{1}{h(d)}\Big), \qquad h(d) = \frac{d}{4}\cot\frac{d}{4},$$

in the frame (axis, two transverse directions), so that

$$\Lambda^J_{\mu\mu} = 1 - \frac{t}{8}\Big[\mu^2 + \frac{J(J+1)-\mu^2}{h(d)}\Big] + O_J(t^2).$$

Here J is fixed and $d \le d_0 < 2\pi$. The exponential of the displayed order-t term is equivalent at this accuracy; noncommuting representation generators prevent treating it as an exact Gaussian characteristic function.

Proof. (a) The geodesic symmetry $g \mapsto m_* g^{-1} m_*$ is an isometry that exchanges the endpoints, so it preserves the bridge law, and in normal coordinates at m_* it is $\xi \mapsto -\xi$. Conjugation by the one-parameter group e^{sX} fixes both endpoints and m_*, so Λ^J commutes with it and is diagonal in the axis basis; the symmetry gives $\Lambda^J = (\Lambda^J)^\dagger$. (b) By (11), off the image terms (which are $O(e^{-c/t})$ for $d \le d_0$), the bridge density is $e^{-(d(e,m)^2 + d(m,e^X)^2)/t}$ times smooth factors. On a space of constant curvature κ, the Hessian of $\frac{1}{2}d(p,\cdot)^2$ at distance r is 1 radially and $r\sqrt{\kappa}\cot(r\sqrt{\kappa})$ transversally. At the midpoint $r = d/2$, $\kappa = \frac{1}{4}$, so the exponent has Hessian $(4/t)\,\mathrm{diag}(1,h,h)$; Laplace's method gives the covariance. Expanding $D^J(e^\xi)$ through second order and using the vanishing odd moments gives the fixed-J order-t formula. □

Correction (GPT-6 Astra referee), 2026-09-27 — REFINE: change the curvature sign, allow weight multiplicities, restrict to a unique minimizing geodesic, and use the Killing form for nonsimple groups.

Any compact connected group, and $SU(3)$. The local covariance argument applies with a bi-invariant metric and X in a compact subset of the injectivity domain. Along its geodesic the curvature operator is $-\frac{1}{4}\,\mathrm{ad}(\hat{X})^2$, with eigenvalue 0 on the centralizer of X and $\frac{1}{4}\alpha(\hat{X})^2$ on the root plane of each positive root α (with $\mathrm{ad}\,X$ having eigenvalues $\pm i\alpha(X)$ there). The midpoint covariance is therefore $t/4$ along the centralizer and $(t/4)/h_\alpha$ on each root plane, with $h_\alpha = \frac{\alpha(X)}{4}\cot\frac{\alpha(X)}{4}$. The centered representation average is Hermitian and commutes with the centralizer of X. It is block diagonal on weight spaces of a maximal torus containing X; weight multiplicities allow mixing within each block. For $SU(2)$ the single positive root has

$\alpha(X) = d$, recovering (b). For a compact simple Lie algebra, $\sum_{\text{all roots}} \alpha(X)^2 = C_2(\text{adj})|X|^2$, and $1/h_\alpha - 1 = \alpha(X)^2/48 + O(\alpha^4)$, the curvature excess averaged over the $\dim G$ directions is, per cut face,

$$\Delta T_j \simeq \frac{t_j}{4} \cdot \frac{C_2(\text{adj})\,|X_j|^2}{48\,\dim G},$$

that is $t_j|X_j|^2/288$ for $SU(2)$ ($C_2(\text{adj}) = 2$, $\dim G = 3$) and $t_j|X_j|^2/512$ for $SU(3)$ ($C_2(\text{adj}) = 3$, $\dim G = 8$, two flat Cartan directions and three root planes).

For a general compact Lie algebra replace $C_2(\text{adj})|X|^2$ by $-\operatorname{tr}(\operatorname{ad} X)^2$; central directions contribute zero and simple factors carry their own metric-dependent Casimirs.

Part (b) gives the leading covariance as $t \to 0$ at fixed X in the specified compact domain. The displayed softening then expands this coefficient at small X. When $|X|^2 \sim t$, other order-t^2 covariance terms also contribute; the softening coefficient alone is insufficient to compute the complete correction at that order. Numerical remainder constants and control after representation summation remain open. Positive curvature enlarges the transverse fluctuation of the midpoint, $1/h(d) = 1 + d^2/48 + O(d^4)$, so the softening of the mid-face weight grows with the side-face flux. Splitting $J(J+1)/h + \mu^2(1 - 1/h)$, each cut face adds $t_j/(4h(d_j))$ to the mid-face heat time, about $t_j d_j^2/192$ above the abelian value $t_j/4$, and an axial term $-\frac{t_j}{8}(1 - \frac{1}{h(d_j)})(\hat{X}_j \cdot T)^2$ of the same order, which couples the mid-face to the direction of the side flux. For fluctuating fields $d_j^2 \sim t$, so both are $O(t)$ relative to the action. N3, the non-Gaussian and commutator corrections in $\prod_j m_j$, enters at the same order and is not computed here. Proposition 7 below gives a weaker finite-cube inequality by bounding all such terms in the remainder; extracting their coupling and irrelevant-operator coefficients requires a separate calculation.

Correction (GPT-6 Astra referee), 2026-09-27 — REFINE: the reflection cancellation is rejected; a conditional finite-cube bound survives with mixed terms in its remainder.

Proposition 7 (conditional finite-cube power bound). Fix a compact connected group, one cube, and positive bounded heat-time ratios. In a local gauge let X collect its independent boundary flux coordinates, $r^2 = \sum_f |X_f|^2$. Assume a uniform Laplace expansion of the *normalized* bridge integral and old-face ratio,

$$\mathcal{D}(X,t) = c(t) + F(X)/t + A(X) - A(0) + R(X,t),$$

with $|F(X) - F_2(X)| \le Cr^3$, $|A(X) - A(0)| \le Cr$, and $|R(X,t)| \le Ct(1 + r^2/t)$. Here F_2/t is the quadratic defect of the isolated Gaussian cube, including the bridge denominators. These bounds require a unique nondegenerate saddle and control of the full heat-kernel/Haar amplitudes. For $r \le C_1 t^{1/2-\delta}$ and $0 < \delta < 1/6$,

$$|\mathcal{D} - c(t) - F_2/t| \le C_2 t^{1/2-\delta}(1 + r^2/t).$$

Thus the unperturbed finite-cube inequality has $\alpha = 1/2 - \delta > 2\delta$, even with coupling shifts set to zero.

Derivation. The cubic classical remainder is bounded by $Cr(r^2/t)$, the amplitude variation by Cr, and $t \le t^{1/2-\delta}$ for small t. This proves the asserted inequality. The following power counting explains the hypotheses without establishing their uniformity. The leading saddle contributions are $S_{\text{cl}} + \frac{1}{2}\log\det H$, where S_{cl} is the minimum over the four midpoints of the exponent $\sum_j [d(Q_j^{-1}, m_j)^2 + d(m_j, P_j)^2]/t_j + d(e, m_{\partial g})^2/(2t_m)$ (the $w = 0$ exponents) and H its Hessian there. One must also add $\sum_j \log k_{t_j}(e^{X_j})$ from the bridge denominators and include the heat-kernel and Haar amplitudes. A remainder order for the bare integral alone does not supply the displayed expansion. (i) *Classical part.* The Lie-algebra linearization of S_{cl} is the abelian problem, so its quadratic part is the free form, which is $\mathcal{D}_0$ after the old-face subtraction. The rest starts at cubic order: Baker–Campbell–Hausdorff terms such as the N1 commutator, $\langle X_{\text{mid}}, [X_{12}, X_{13}]\rangle/t$, and quartic invariants $|X|^4/t$. On the small-field set, $|X|^3/t \le t^{1/2-\delta}|X|^2/t$

and $|X|^4/t \le t^{1-2\delta}|X|^2/t$, both inside the remainder of $\mathrm{P}(\alpha)$ with $\alpha = \frac{1}{2} - \delta$. (ii) *One-loop part.* Mixed quadratic boundary terms can survive cube reflections because reflections also permute the faces. For example, $\langle \Delta_1 X_{23}, \Delta_2 X_{31} \rangle$, with Δ_i the difference between opposite faces normal to direction i, is even under each coordinate reflection: each factor transforms as a pseudoscalar. Axis permutations can be accommodated by summing the corresponding pairs. The linear Bianchi relation among the three differences can rewrite such terms as squares of differences, which still couple opposite faces. Thus reflection invariance permits more than a sum of individual plaquette norms. The coupled Hessian can also have linear off-diagonal blocks from cubic BCH terms, whose squared contribution enters $\log\det H$ at quadratic order. Every bounded quadratic amplitude term is $O(r^2) = O(t)(r^2/t)$ and already fits the stated remainder since $\alpha < 1$. Identifying plane-diagonal coupling shifts requires a separate calculation. The present bound makes no such identification. (iii) The old-face ratio k_{2t}/k_t is even in $\log U_f$ and contributes a constant and a transverse coupling shift. □

Proposition 7 is power counting made explicit for one cube; its hypothesis, a uniform Laplace remainder on the small-field set, is exactly the missing estimate. Proposition 6 computes the curvature part of (ii) for $SU(2)$. For a whole mid-plane the same bookkeeping applies, but H couples all mid-edges and $\log\det H$ is non-local in the fluxes; controlling it is the cluster expansion of open problem 2 in the README.

Under $\mathrm{P}(\alpha)$ the per-step error on smooth observables is $O((\lambda_3 a)^\alpha)$ and the coupling shifts sum to a finite renormalization, because $\sum_n \lambda_3 a_n = 2\lambda_3 a_0$: the superrenormalizable pattern of the refinement note's table, now with the term to be bounded identified exactly. $\mathrm{P}(\alpha)$ is the directional form of what Balaban's three-dimensional ultraviolet stability controls in his block-averaging scheme (Balaban 1985, CMP 102, 255, metadata); whether his estimates transfer to the heat-kernel directional scheme is unverified. $\mathrm{P}(\alpha)$ yields ultraviolet control and says nothing about the gap: the long-distance spectral estimate of the refinement note §7 remains a separate obligation.

5. Four dimensions

In $D = 4$ a halving of direction 1 inserts parallel copies in the three planes $(23), (24), (34)$, and the mid-space is a three-dimensional lattice gauge theory with bridge-distributed edges. By the table in §2 the heat time returns to itself after an isotropic step, so the whole one-loop running $g_0^{-2}(2a) = g_0^{-2}(a) - 2b_0 \log 2 + O(g_0^2)$, with $b_0 = 11/(16\pi^2)$ for $SU(3)$ in the convention $g^2_{\mathrm{lat}} = \lambda_4$, must come out of the four Ψ factors of one isotropic step, to leading order and within the class of actions of $\mathrm{P}(\alpha)$. N2 and N3 are $O(t)$ relative in both dimensions. In $D = 4$, $t = g^2$ does not decrease along the sequence, so their step-independent shift of g^{-2} is the whole one-loop running, $-2b_0 \log 2$ per isotropic step; in $D = 3$ the shifts, $O(\lambda_3 a_n)$ relative, are summable. This is the exact location of marginality in the insertion language. The corresponding hypothesis is H1 of the conditional theorem, and Balaban's four-dimensional work (Balaban 1987, metadata) is the established partial control.

6. Consequence for STATE

Item 1 of STATE is answered in its "exact obstruction" form, and its first rungs are proved. One directional refinement step factors exactly into series moves, which close, and a parallel insertion Ψ (Proposition 1). Ψ is exact for the free field (Proposition 2), for the $U(1)$ cube (Proposition 3) and the $U(1)$ mid-plane (Proposition 4), and for $U(1)$ the unperturbed one-step bound has error density $e^{-\pi^2/(8\lambda_3 a)}$ (Theorem 5). For non-abelian groups the midpoint curvature term is explicit (Proposition 6 and its group-general form), and the finite-cube bound is conditional on a full normalized Laplace expansion (Proposition 7). The next step is the uniform Laplace remainder that turns Proposition 7 into a theorem for $SU(2)$ and $SU(3)$; after it, a mid-plane by small-field cluster expansion, and stability under iteration. For the paper, the series/parallel

split is the organizing statement of the dimension comparison: Newton's time insertion is a series move closed by one scalar; two-dimensional gauge theory has only series moves; three and four dimensions add $\binom{D-1}{2}$ parallel insertions whose elimination is the renormalization problem; the error density of those insertions in physical units gives a sufficient test for which lattice effects the limit discards.